



► RoboSim

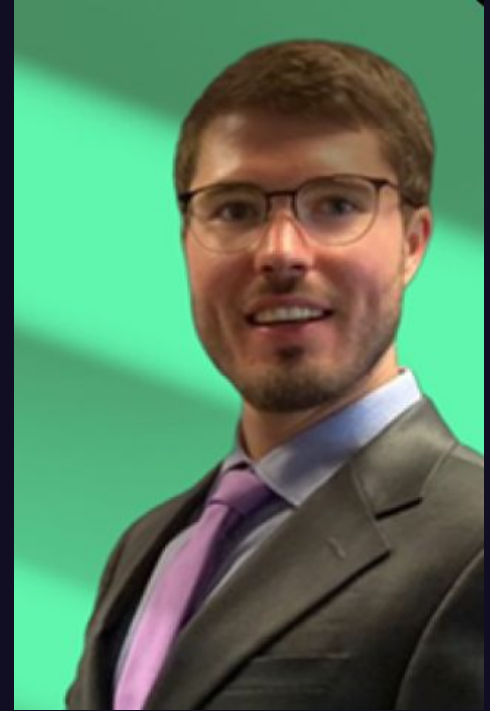
Project team : Freya, Cassie, Raymond, Mincheol, Ziyang

► OUR PROJECT

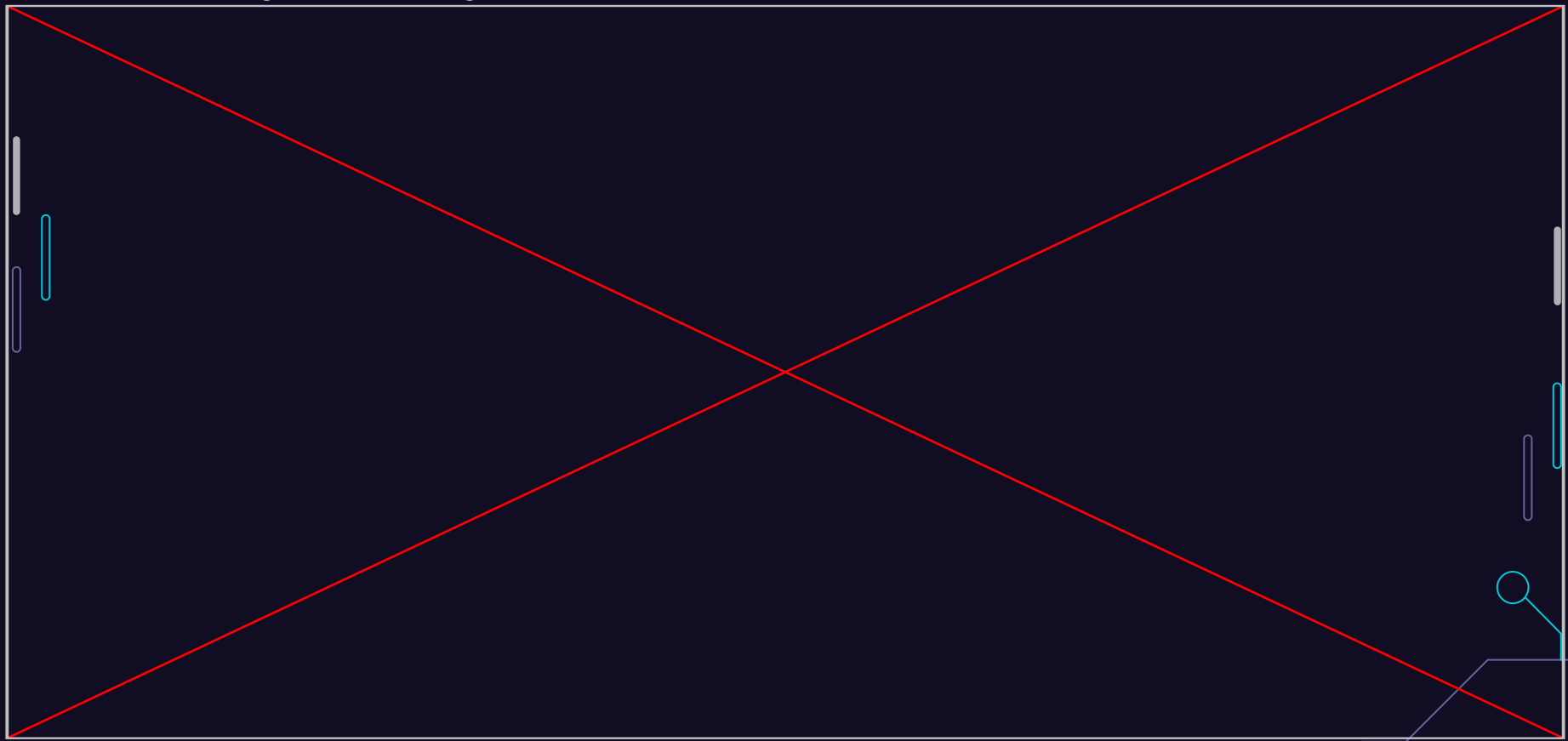
- Marius Jurt: Masters Student at Bristol, Research Engineer at Toshiba.
- RoboSim used to be a Toshiba project which Marius took on.
- Self-moving, self-stacking robotic boxes.
- RoboSim is a simulation platform for algorithm developers.
- Ranks all algorithms on leaderboard based on time/moves.



Robot boxes at MIT climbing up their neighbours



► Last year's system



► USER STORIES



WAREHOUSE MANAGER

I want to optimise storage and retrieval of goods, so that I can increase efficiency to reduce the overheads of inventory management.



ALGORITHM DESIGNER

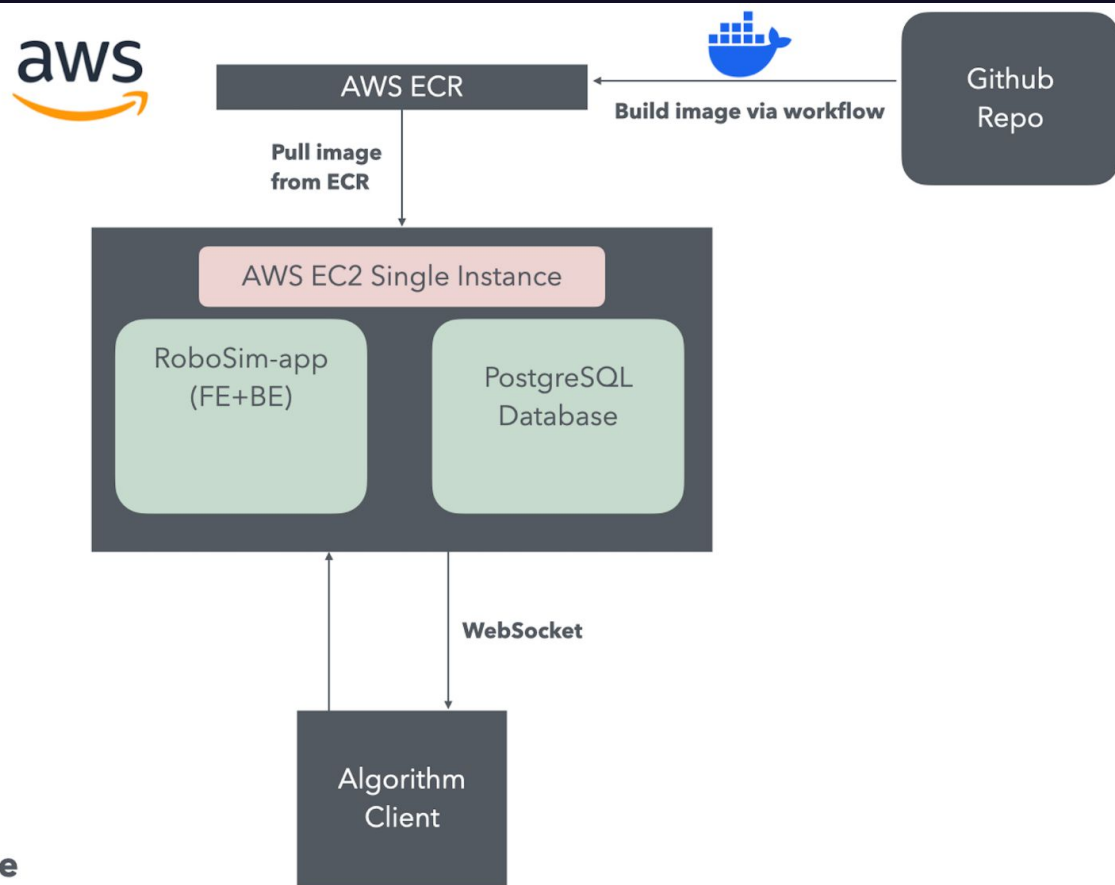
I want to develop and benchmark routing algorithms with an intuitive API, so that I can streamline testing and optimisation of robot navigation in 3D space.



FUTURE CONTRIBUTOR

I want support with understanding the existing codebase and project.

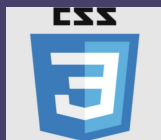
ARCHITECTURE DIAGRAM



Our Project Architecture

Tech Stack

Web Client:



Ts TypeScript



Database:



Server:

Ts TypeScript



Algorithm Client:



► DEVELOPMENT



DEVELOPMENT

We had some problems with code (e.g. the algorithm client) not working on some people's machines.

To fix this, we plan to start implementing a CI system, and hopefully CD in the future.



USER TESTING

Last year's team made unit tests for the code.

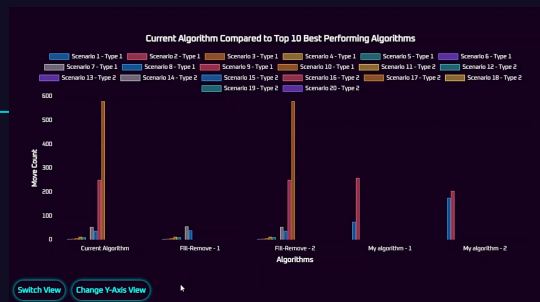
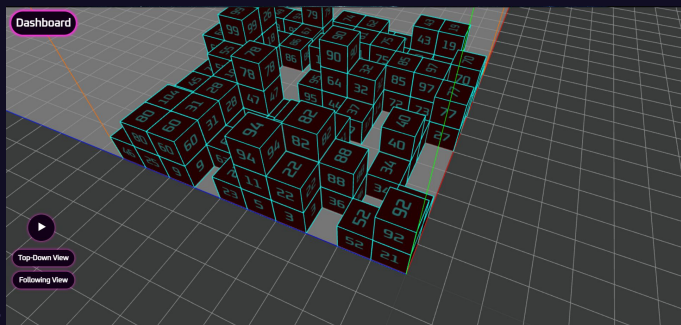
We plan to build upon these tests in the future so that we can monitor and improve code coverage.

► INTERFACE DESIGN/FUNCTION



FEATURES

- Review each run with the 3D warehouse replay.
- Visualise benchmarking results (time / moves) with graphs and tables.
- Use the playground to build custom scenarios.



USERS

- See whether each algorithm run passed or failed its scenarios.
- Inspect how boxes moved step-by-step in the 3D replay.



UPDATES

- Added perspective view modes (top-down / follow box).
- Prepared UI hooks for replay controls (start / pause)
- Planned: path-tracing of the travelling box and a predicted energy-cost metric integrated into the leaderboard.

► CLIENT FEEDBACK



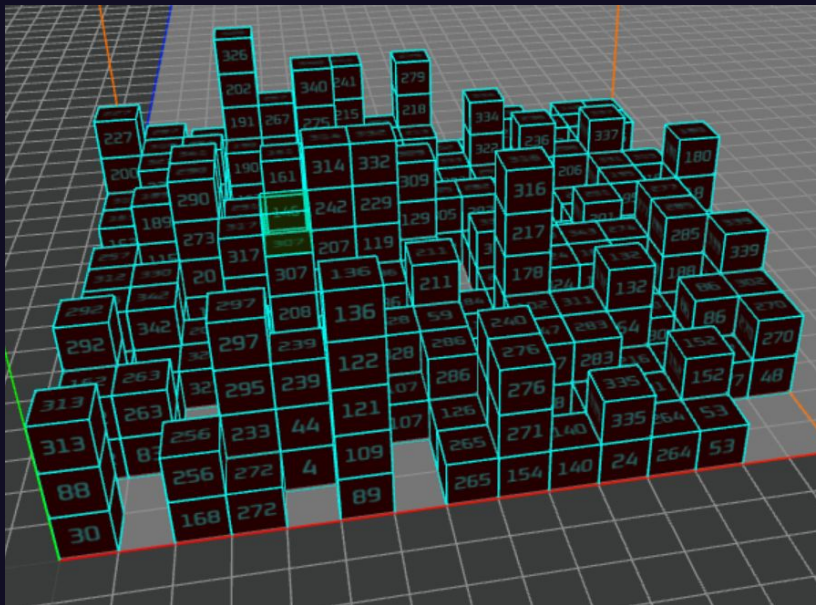
DEPLOYING SYSTEM

Marius encouraged us to deploy the system as early on as possible.



REAL WORLD CONSTRAINTS

- Realistic warehouse.
- Type 2 boxes - how many can it hold on-top?
- Benchmarking: stats for target box moves to other boxes moves.



► PROJECT MANAGEMENT



MEETINGS

Once weekly before our mentor meeting, changed to meeting our mentor first for more direction



COLLABORATION

Some joint programming on the database, we met up to merge all the branches the weekend before MVP



WEEKLY TASKS

In meetings we updated each other on what we'd been doing and set ourselves tasks for the next week so everyone had something to do

► CHALLENGES

- **Database:** initial libraries weren't suitable, syntax error with pre-existing code, asynchronous issues, too many files accessing it.
- **Pre-existing codebase:** daunting at first and took a long time to understand. And getting the existing code set up on all our laptops.
- **Requirements:** initial client meeting didn't give much direction, so we contacted him again for more specific requirements.
- **Commits:** Some people weren't committing frequently - doing big commits instead, this improved towards the deadline after communication.

A screenshot of a login form titled "RoboSim Login" in a bold, pink font. The form is set against a dark blue background with a glowing pink border. It contains two input fields: "Username:" with a light blue box containing the text "username", and "Password:" with a light blue box containing eight dots. Below these is a pink "Login" button. Under the button, there is a red error message "Invalid Username or Password" and a link "Do not have an account? Register now!" in white text.

RoboSim Login

Username:

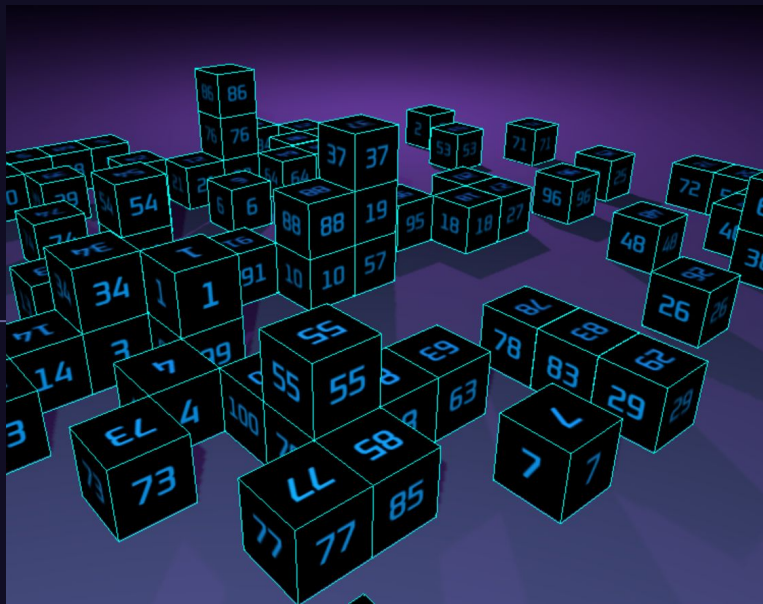
Password:

Login

Invalid Username or Password

Do not have an account? Register now!

► OUR PLAN



- **Rotating roles:** means we all get a go to experience each side of the development.
- **Deploy system:** one of our main feedback points from Marius.
- **Implement CI/CD**
- **Improved Type 2 algorithm**
- **More features:** we have many ideas and asked Marius what features he'd like us to consider.



► **DEMO!!**